



Motor System - Part 2

S. Bapi Raju

[raju.bapi<at>iiit.ac.in](mailto:raju.bapi@iiit.ac.in)

Recap

- Memory taxonomy
 - Sensory, STM and **LTM**
 - LTM
 - Explicit (Declarative) vs **Implicit (Non-declarative)**
 - **Procedural Memories** are motor programs
- Procedural memories are special
 - difficult to acquire but hard to forget!
- Topographic organization (homunculus)
 - Primary organs of movement in M1
 - Ex.: legs medially, hands laterally, etc

Recap

- Key Brain areas involved in action or movement:
 - **Subcortical areas:**
 - **Basal Ganglia (BG):** Initiation, voluntary control, RL
 - **Cerebellum (CB):** Coordination
 - **Cortical areas:**
 - Motor Cortex (Primary, Secondary, Supplementary)
 - Planning and execution of voluntary movements
- Motor Control:
 - **open loop** (no feedback incorporated)
 - **closed loop** (feedback incorporated)
- Handedness inventory and theories

Clinical (Parkinsonian)

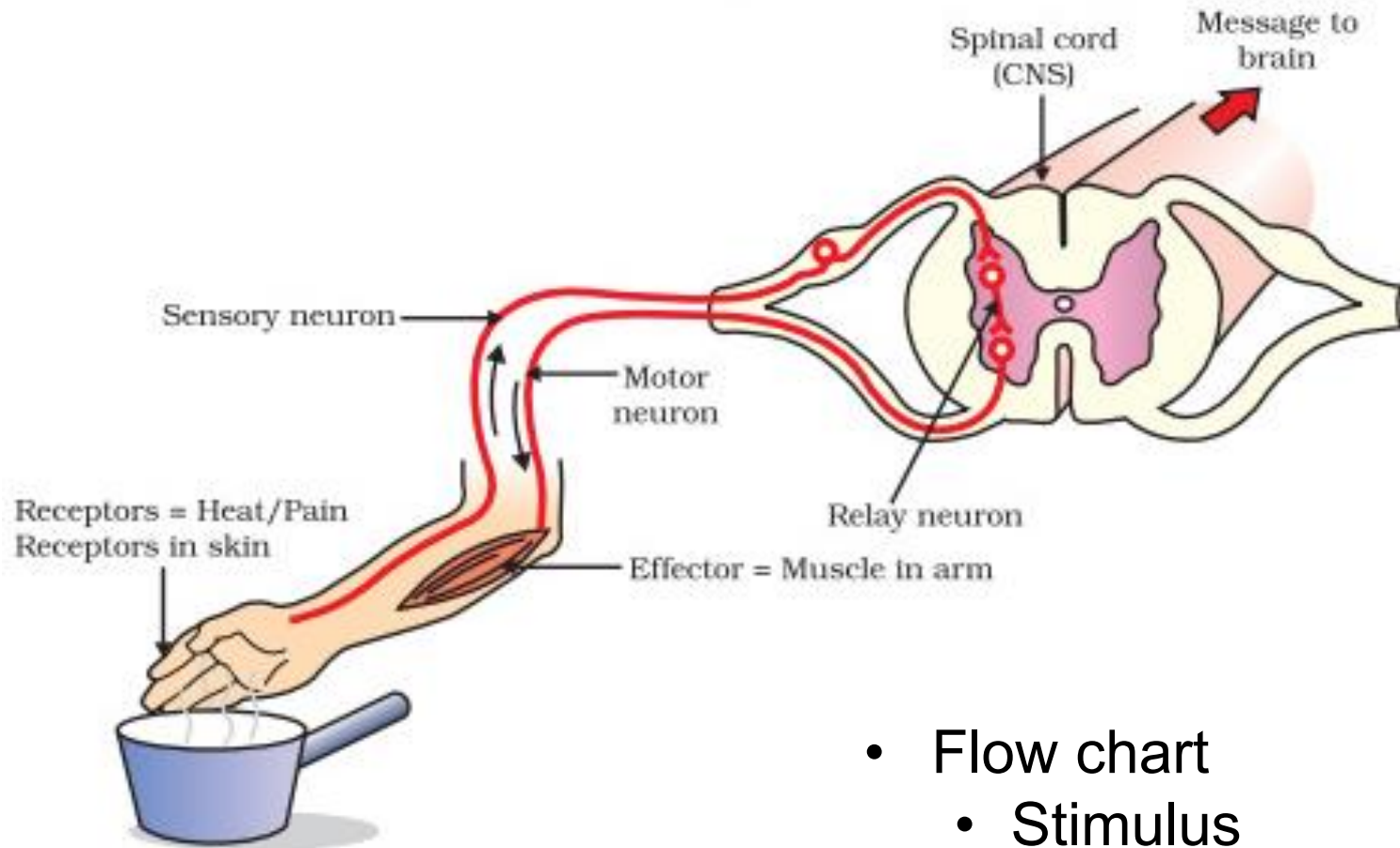
November 13, 2010 was my (Kenny) one-year anniversary. I had bi-lateral Deep Brain Stimulation surgery (DBS) at Stanford University Medical Center on this date in 2009. With the stimulator turned OFF, the Parkinson's symptoms return with a vengeance.

Watch how a few milliamps of current can change a person's life.

check out this follow up video:

www.youtube.com/watch?v=17ch1guvoLA

Reflex Arc

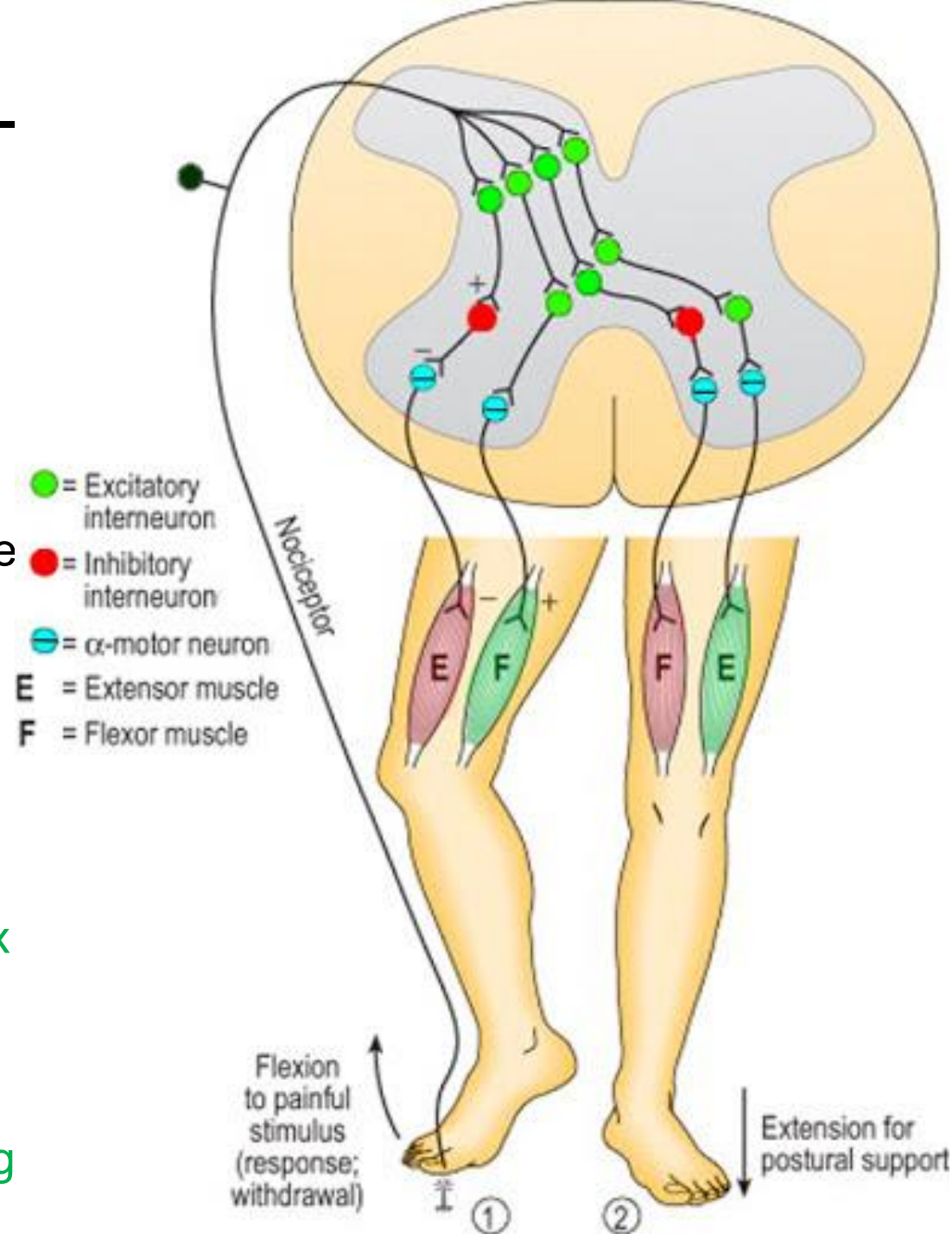


A **reflex** is a fast, predictable, automatic (involuntary) response to a stimulus (change in the environment)

- Flow chart
 - Stimulus
 - Sensory Neuron
 - Interneuron (Relay neuron)
 - Motor Neuron
 - Response

Complex Reflex

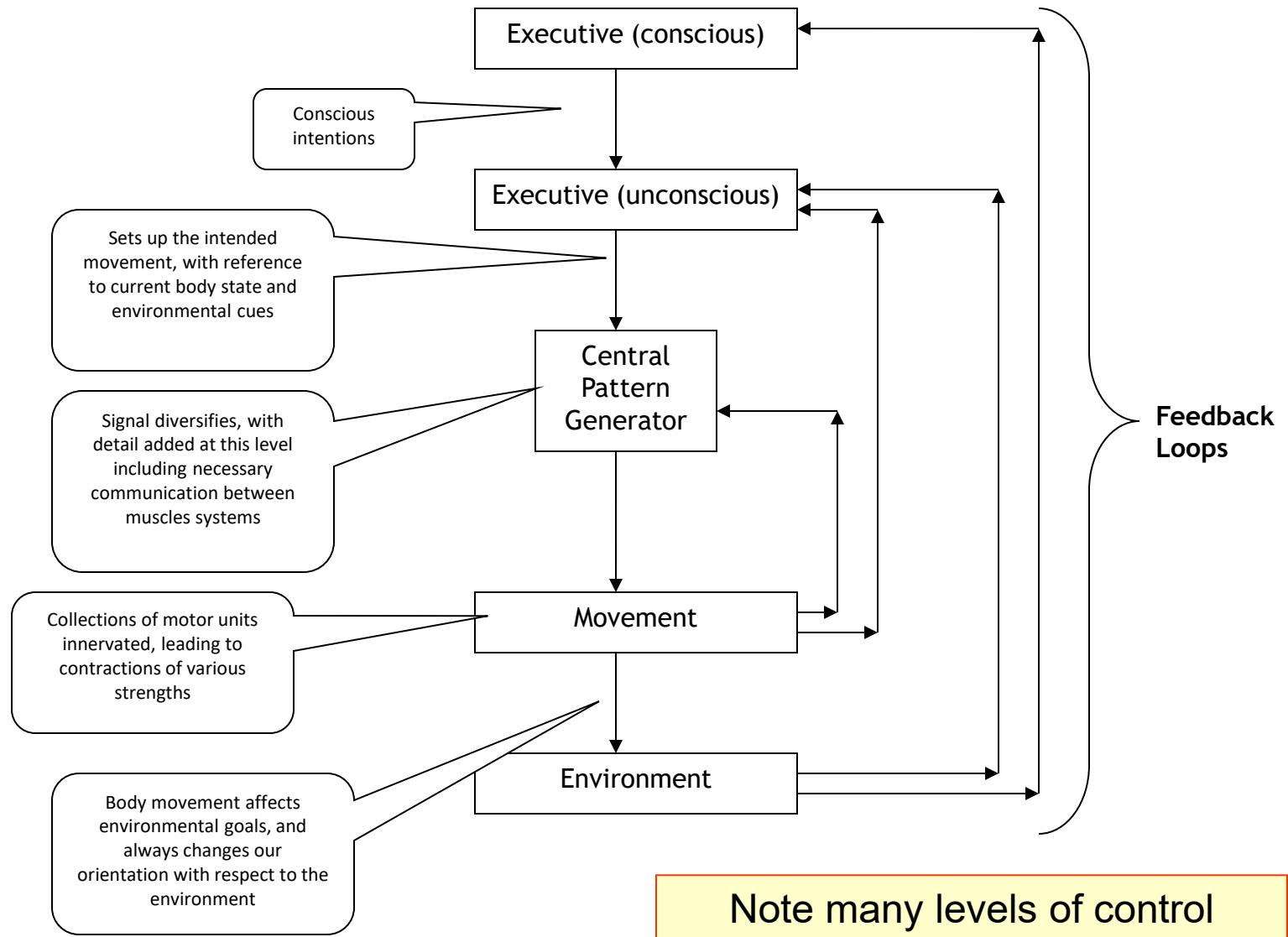
- **Flexor and Crossed Extensor Reflex**
(Sheridan 1900)
 - flexors in the withdrawing limb contract and the extensors relax, while in the other limb, the opposite occurs. The crossed extensor reflex is contralateral, meaning the reflex occurs on the opposite side of the body from the stimulus
 - **Example: withdrawal reflex of leg when a nail pierces the foot and posture stabilization with other leg supporting by straightening**



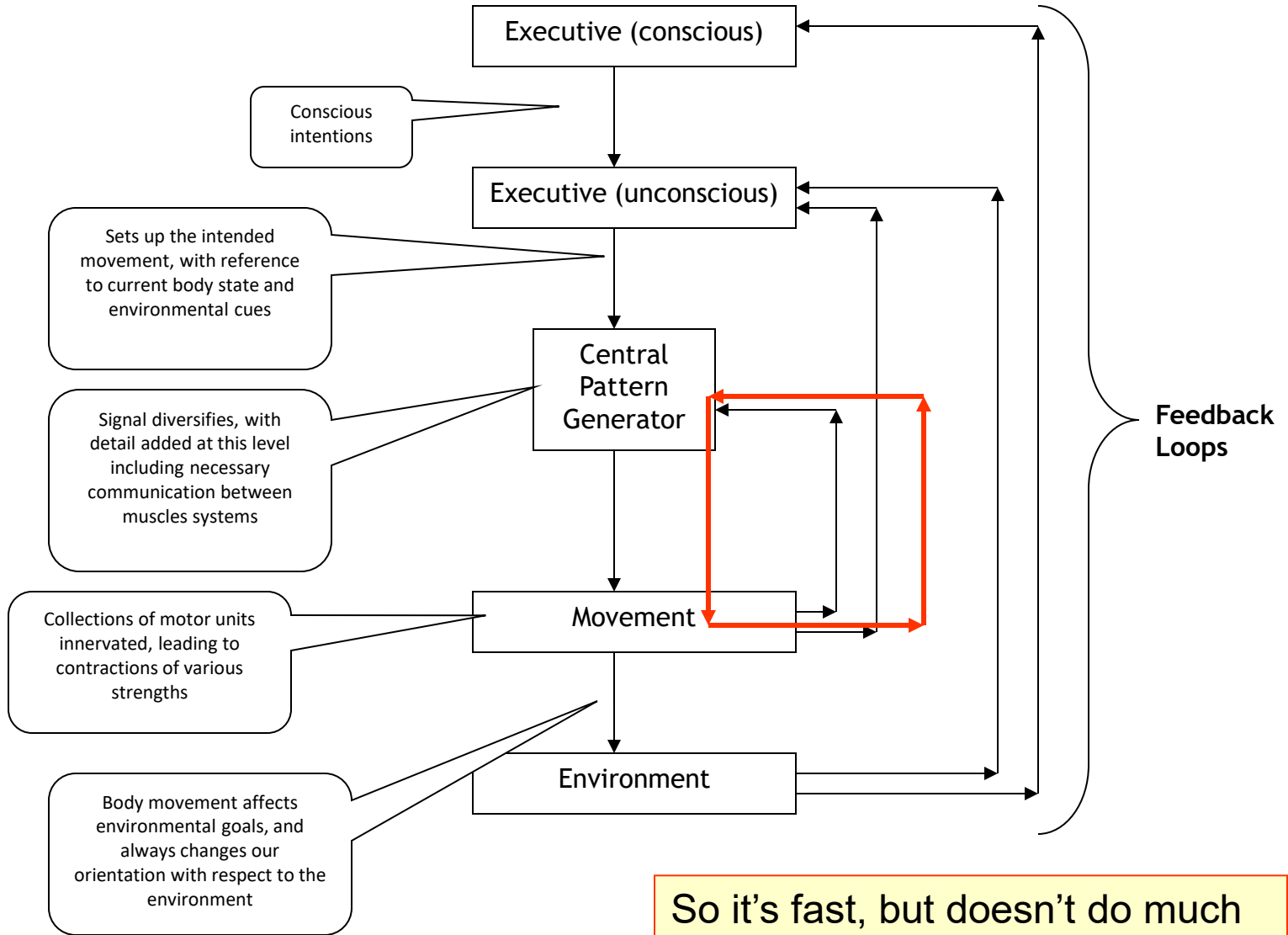
Hierarchy of Motor Control

LEVEL	FUNCTION	STRUCTURES
high	strategy	association areas of neocortex, basal ganglia
middle	tactics	motor cortex, cerebellum
low	execution	brain stem, spinal cord

1. An Information Processing model of Hierarchical Motor Control

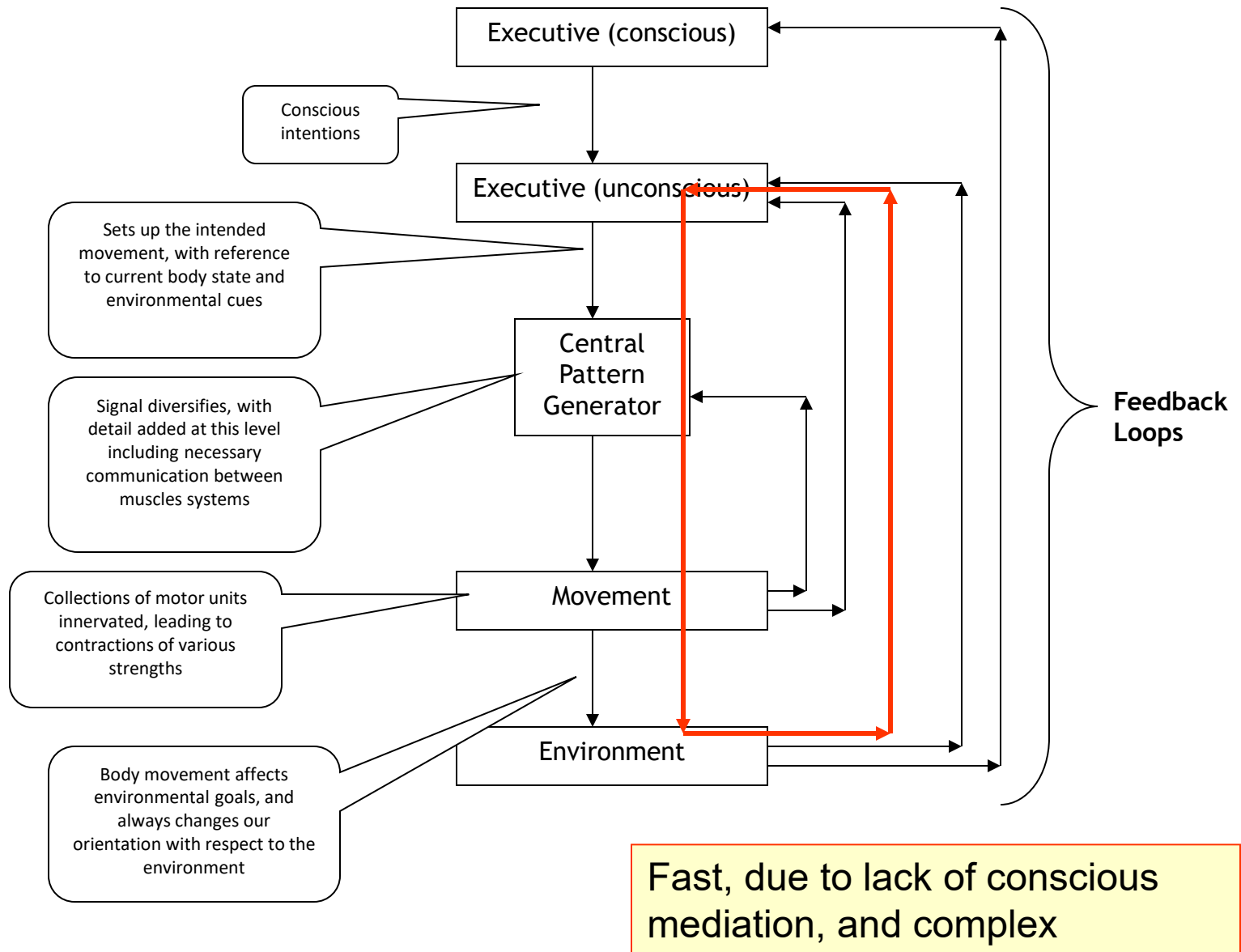


2. A knee-jerk (stretch, myotatic, monosynaptic) reflex

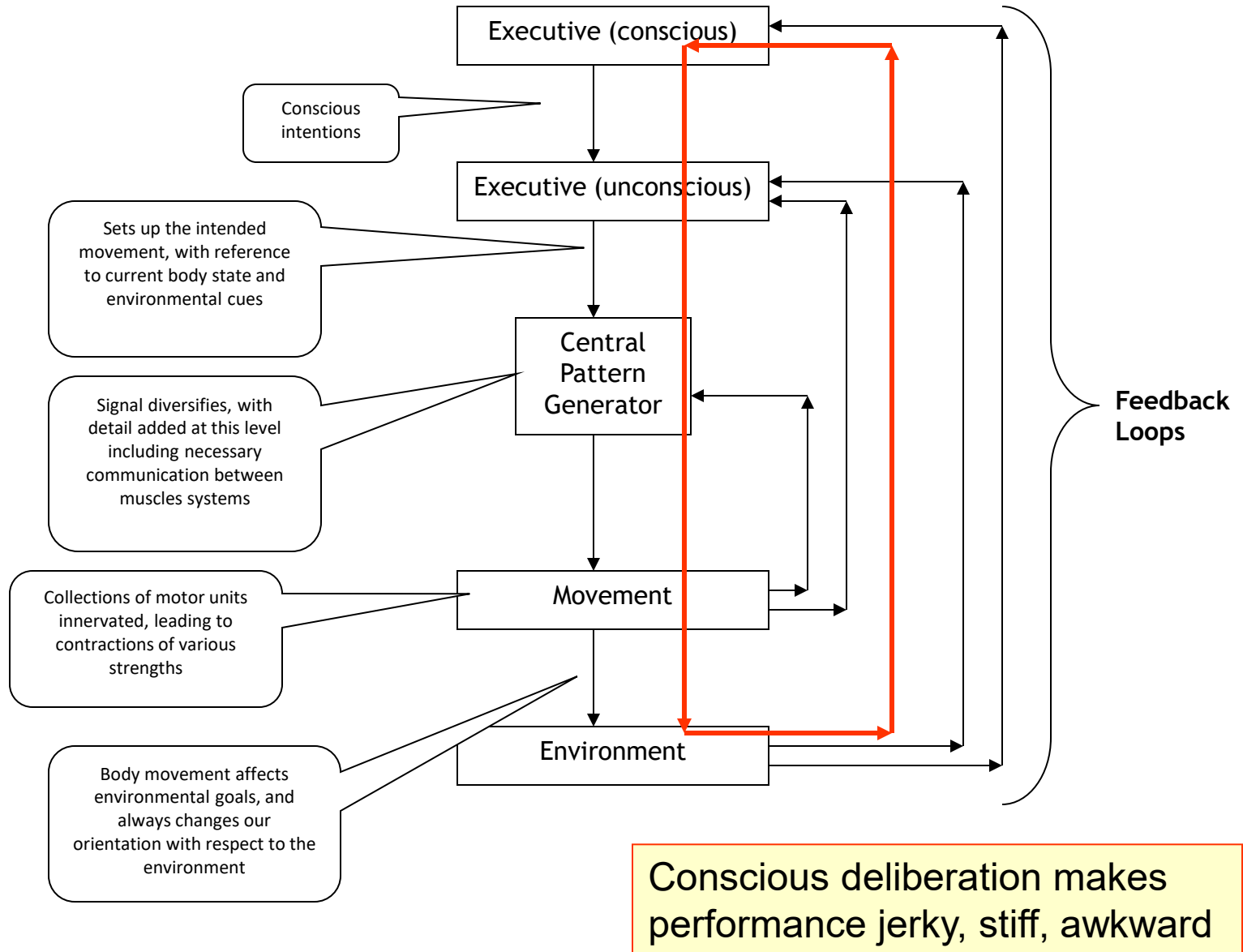


So it's fast, but doesn't do much in terms of coordination

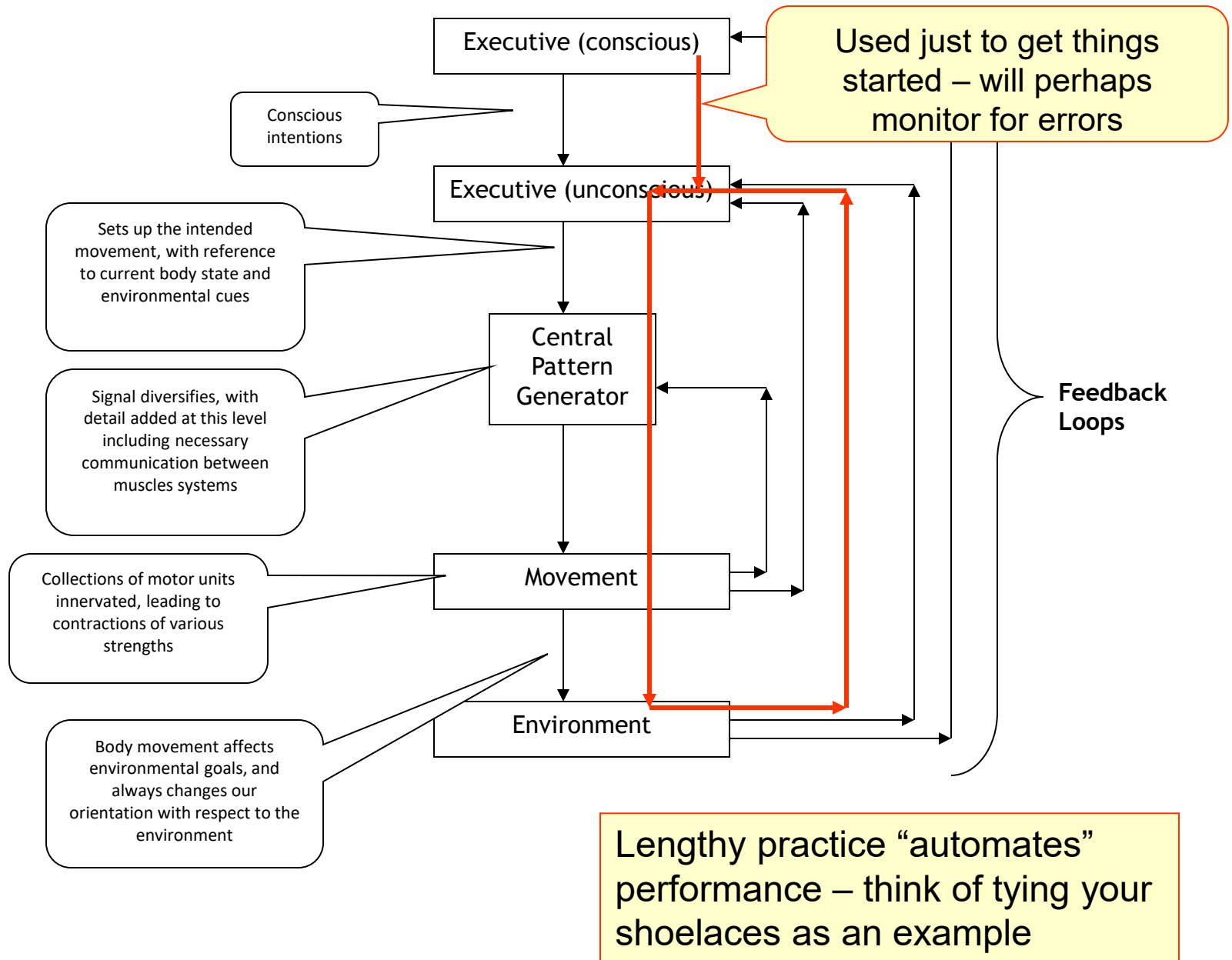
3. Balance (Swinging room and Normal



4. Poorly learned skill



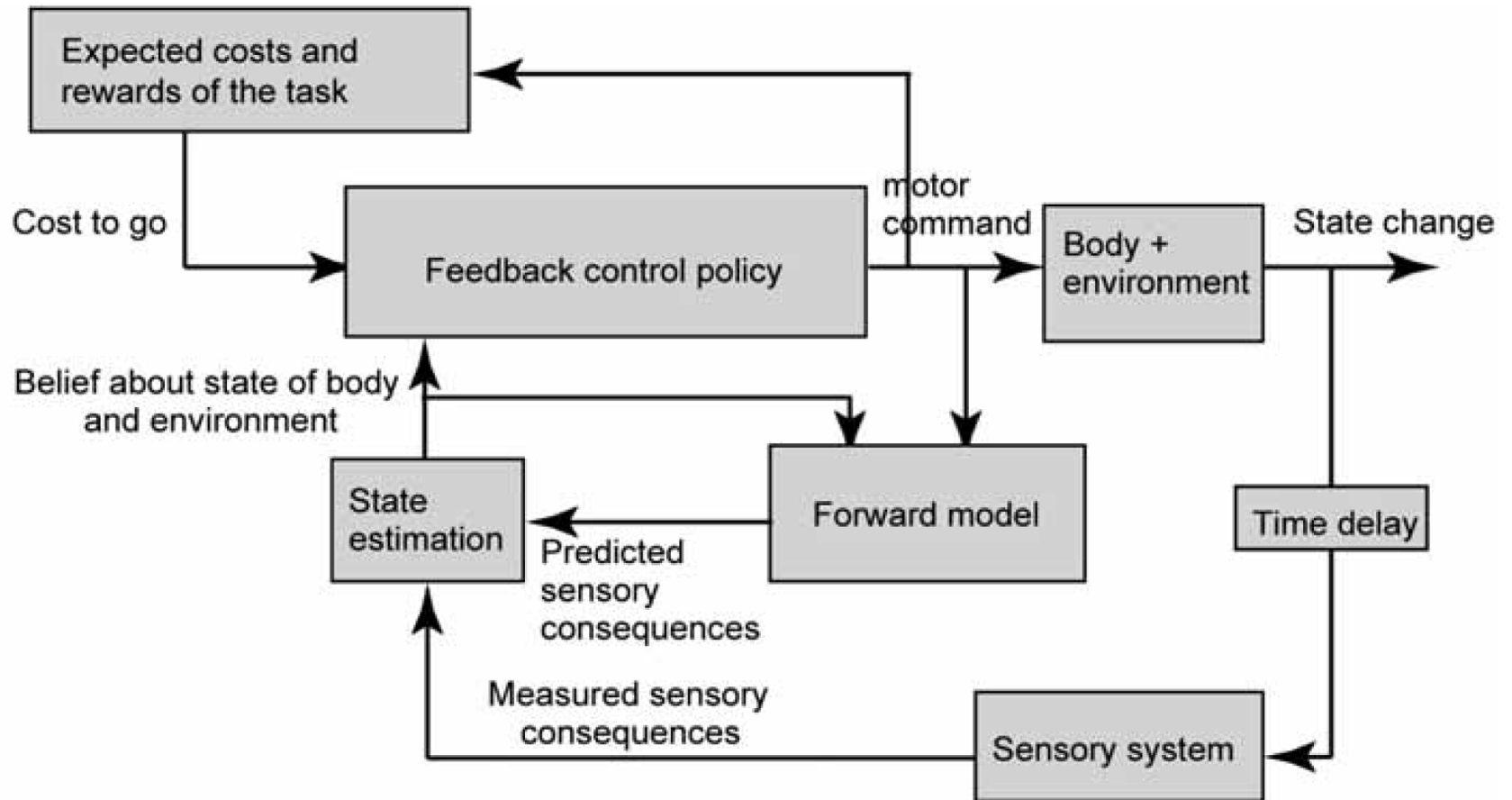
5. Expertly learned skill



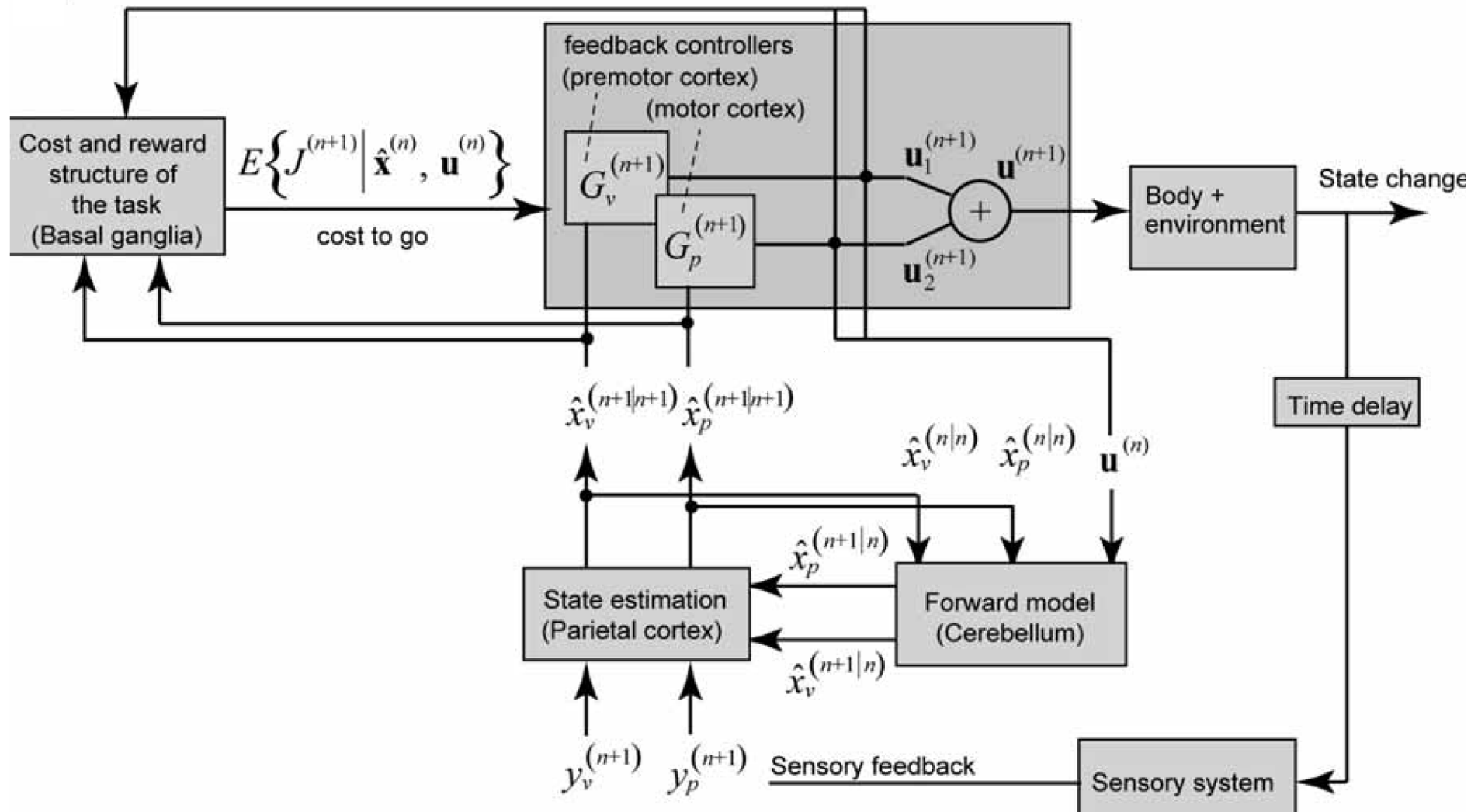
Goal-Directed Behaviour

- Humans move with a purpose. How does the nervous system enable **goal-directed behaviour**?
 - The intrinsic neural codes that are used to generate motor actions are not yet fully known.
 - The principles that lead to optimal performance are only just beginning to be modeled.
 - The system is **non-stationary** in that adaptation and skill learning appear continuously and on many time scales.

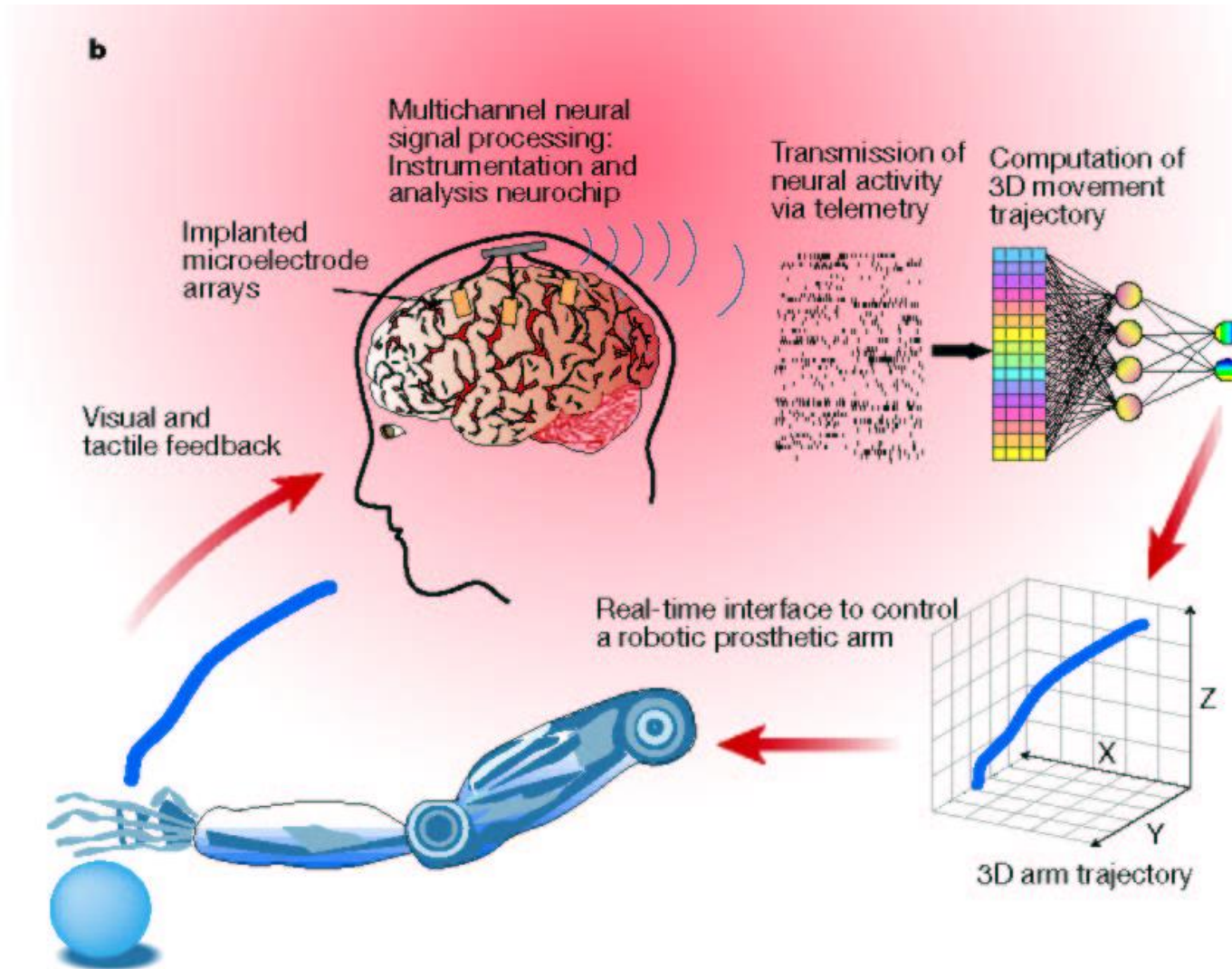
Goal-directed movements



Goal-directed movements



Prosthetics



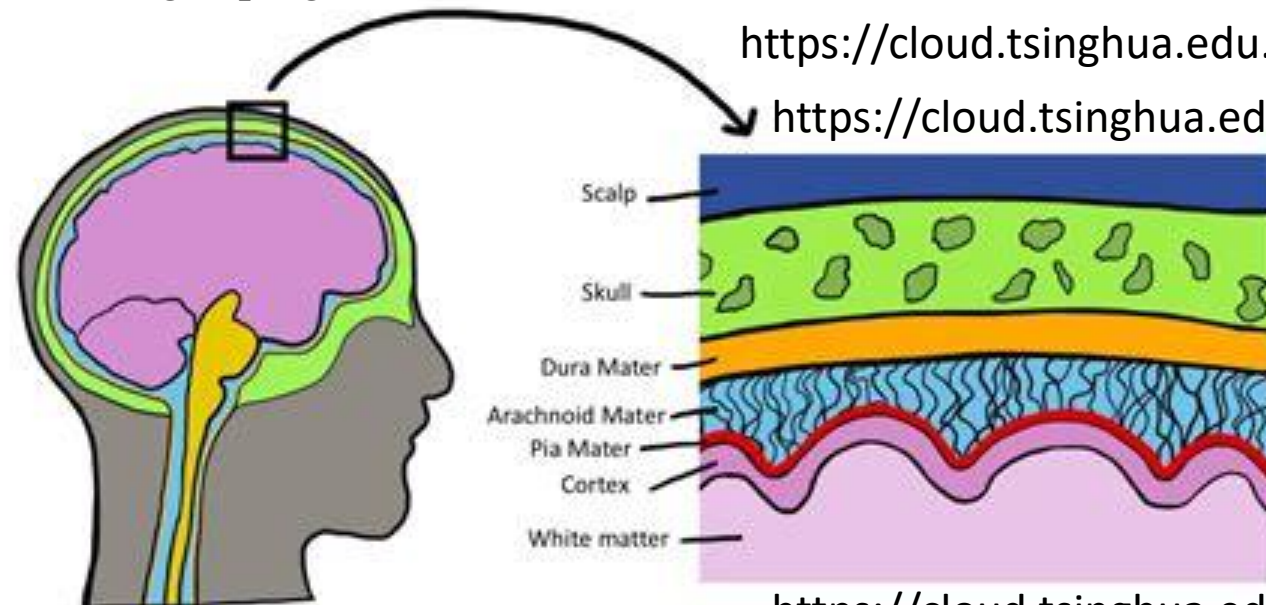
- Engineering application: cortically-controlled neural prosthetics

Neo BCI: Neuracle & Tsinghua, China 2025

- The coin-sized NEO (minimally invasive BCI) is embedded in the skull, with eight chronic epidural electrodes placed over one side of the primary sensorimotor cortex to record electrical activity when the person imagines moving their opposite hand. The signals are sent to a computer to be decoded and are then used to control a glove, allowing people to pick up and move objects to perform daily functions such as eating and drinking.
- A complete spinal cord injury (SCI) patient with paralyzed hand functions was recruited in this study. Over a 9-month period of home use, the patient achieved an average grasping detection F1-score of 0.91, and a 100% success rate in object transfer tests.

<https://cloud.tsinghua.edu.cn/f/ba945e673ec74af5a2b5/>

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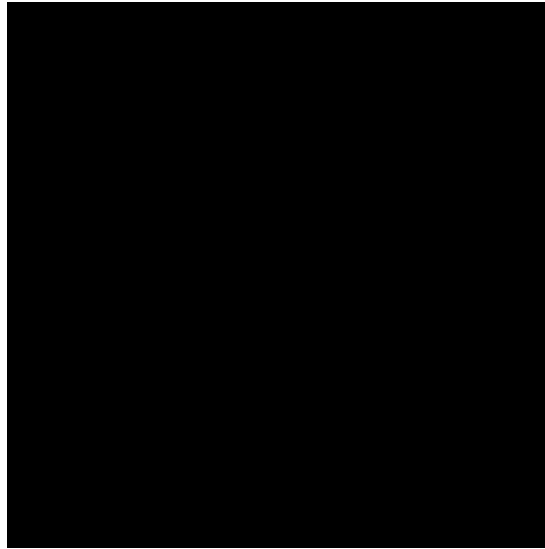
Liu, D. *et al.* Reclaiming Hand Functions after Complete Spinal Cord Injury with Epidural BCI.

medRxiv <https://doi.org/10.1101/2024.09.05.24313041> (2025)

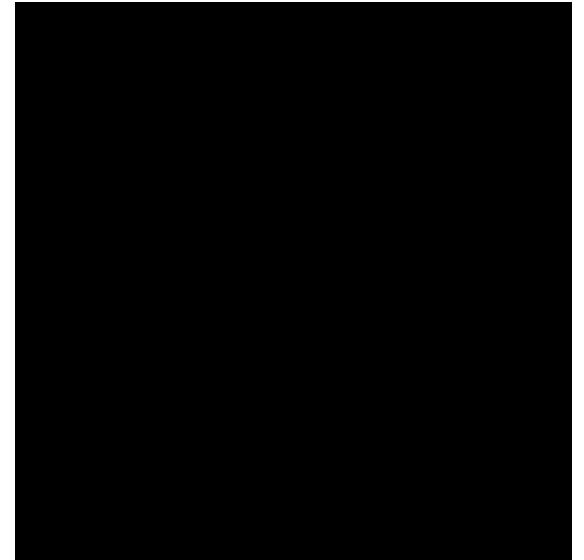
Skill Learning



Cognitive



Associative



Automatic

- 3 Phases of Skill Learning (URL: <https://youtu.be/n7UcobScnck>)

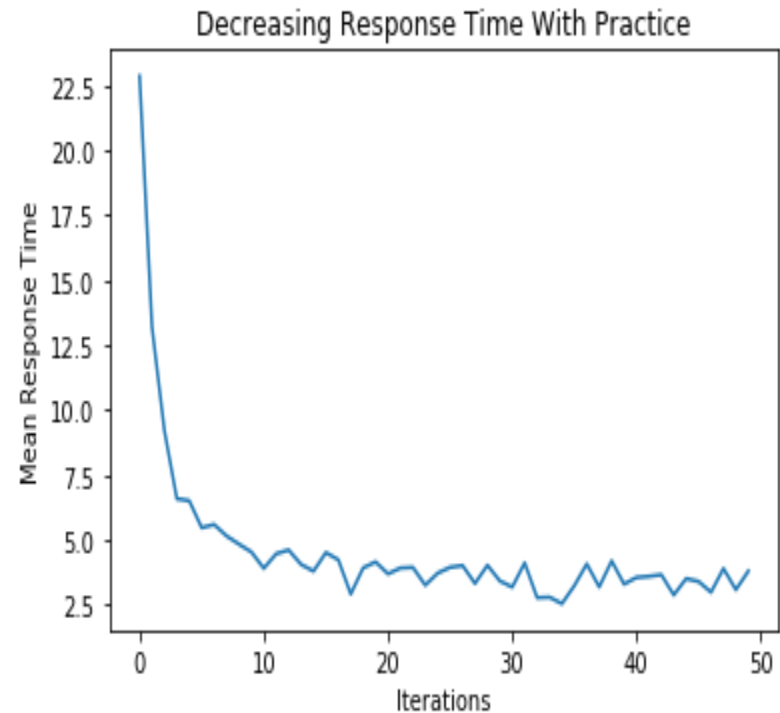
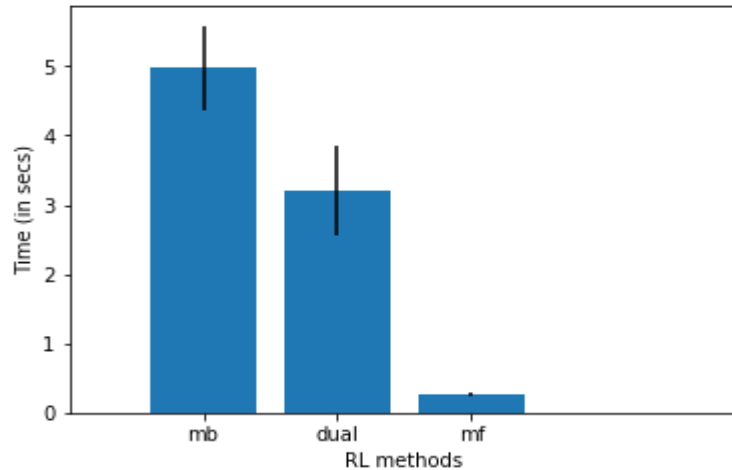
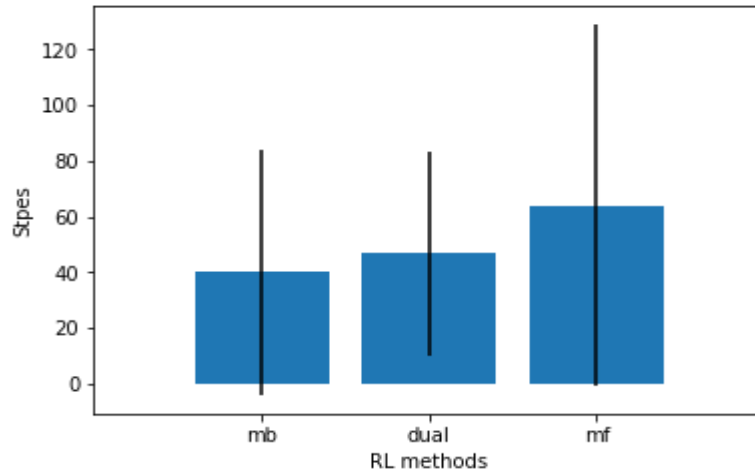
Skill Learning: Fitts' 3 Phases

Stages of Learning	Characteristics	Attentional Demands
Cognitive Phase (Verbal)	Movements are slow, inconsistent, and inefficient	Large parts of the movement are controlled consciously deliberately
	Considerable cognitive activity is required	
Associative Phase	Movements are more fluid, reliable, and efficient	Some parts of the movement are controlled consciously, some automatically
	Less cognitive activity is required	
Autonomous Phase (Motor)	Movements are accurate, consistent, and efficient	Movement is largely controlled automatically
	Little or no cognitive activity is required	

Research Questions

- I. How can we investigate skill learning?
 - Empirical Studies:
 - Design Canonical tasks, investigate behavior, Neuroimaging
- II. What computations underlie different phases?
 - Reinforcement Learning (RL) Models
 - Skill learning = Sequential Decision Making
 - Model-Free and Model-Based RL
- III. How does “agency / self” arise in voluntary motor control?

Dual-Process Model: Model-Based + Model-Free RL



- Dual Process Proposal: Savalia et al., (2016) Frontiers in Psychology